

DM IMCQ 4

Spring 2026

Physiology, Histology, Anatomy, Biochemistry

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KEY file

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Physiology 2 Questions

Investigators conduct a study examining the mechanism of acid secretion by stomach during gastric phase of digestion. For this study, a group of 50 healthy men between 20–30-year age is selected. A balloon is inserted into the stomach of these participants. The stomach is distended by inflating the balloon for 30 minutes. It is observed that the serum gastrin level increased through a vago-vagal reflex. Release of which of the following neurotransmitters from the vagus nerve are most likely responsible for this serum alteration seen in the participants?

- A. Histamine and acetylcholine
- B. Gastrin-releasing peptide (GRP) and acetylcholine
- C. Somatostatin and histamine
- D. Histamine and Gastrin-releasing peptide (GRP)
- E. Vasoactive intestinal polypeptide (VIP) and acetylcholine

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A 73-year-old man comes to the clinic because of a 2-month history of abdominal discomfort, intermittent diarrhea, and a 3-kg (6.6-lb) weight loss. Physical examination shows no abnormalities. Laboratory studies show a serum somatostatin concentration of 360 pg/mL (Normal:<25 pg/mL). MRI of the abdomen shows a tumor mass in the pancreas as shown with arrows. He is diagnosed with somatostatinoma (a tumor hypersecreting somatostatin). Which of the following sets of gastric functional alterations is most likely in this patient?



Option	Gastric pH	Gastrin secretion	Histamine secretion
A	↑	↑	↑
B	↑	↑	↓
C	↑	↓	↓
D	↓	↑	↑
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D	↓	↑	↑
E	↓	↓	↓

SOM.MK.II.BPM2.3.DM.1.P
HYS.1098. Compare and contrast the regulation of gut function by nerves, hormones, and paracrine regulators.

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Histology 2 Questions

A 55-year-old woman is brought to the emergency department because of a 6-week history of fatigue, generalized weakness, and dizziness. Her blood pressure is 100/60 mm Hg and pulse is 100/min. Physical examination shows pale conjunctivae and oral mucosa. Laboratory studies show a vitamin B12 concentration of 175 ng/mL (normal range 200-900 ng/mL). Pernicious anemia is suspected, and an upper GI endoscopy with gastric biopsy is scheduled. Which of the following is most likely a biopsy finding in this patient?

- A. Atrophy of Paneth cells
- B. Hypoplasia of mucus cells
- C. Under-development of goblet cells
- D. Hyperplasia of chief cells
- E. Destruction of parietal cells

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A 67-year-old man is brought to the emergency department because of a 1-hour history of vomiting of bright red blood. He has a history of Alcohol Use Disorder. His blood pressure is 90/60 mm Hg, and pulse is 110/min. Investigations show signs of portal hypertension. Upper gastrointestinal bleeding because of the rupture of dilated esophageal veins (esophageal varices) is suspected. Which of the following is most likely a normal histological feature of the layer of the esophageal wall in which the bleeding vessels are normally located?

- A. Stratified squamous epithelium
- B. Myenteric plexus
- C. Mucus glands
- D. Cardiac glands
- E. Smooth muscle bands

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State the layers of the wall of the esophagus.SOM.MK.I.BPM2.3.DM.1.HCB.1204

List the components found in the layers of the wall of the esophagus.SOM.MK.I.BPM2.3.DM.1.HCB.1205

Describe the microscopic structure of the esophagus.SOM.MK.I.BPM2.3.DM.1.HCB.1206

Differentiate between esophageal cardiac and proper glands based on their location and type of

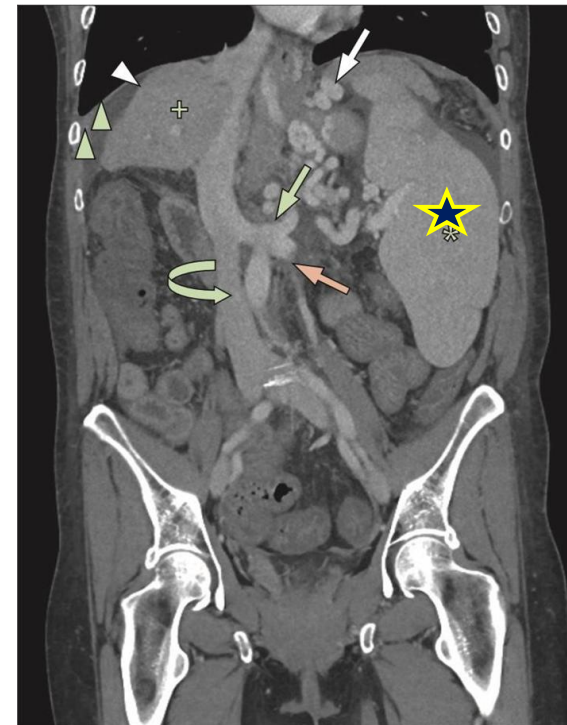
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Anatomy 3 Questions

A 64-year-old man comes to the physician because of dark sticky stool for the past few weeks. He has a chronic history of liver cirrhosis and an extensive alcohol use. He appears malnourished. Physical examination shows engorged veins around the umbilicus and organomegaly. Imaging studies show cirrhosis of the liver and enlargement of the highlighted organ (see star in attached image). A diagnosis of portal hypertension is made. Hypertension of the vein draining the enlarged organ will most likely result in congestion of blood in which of the following vessels?

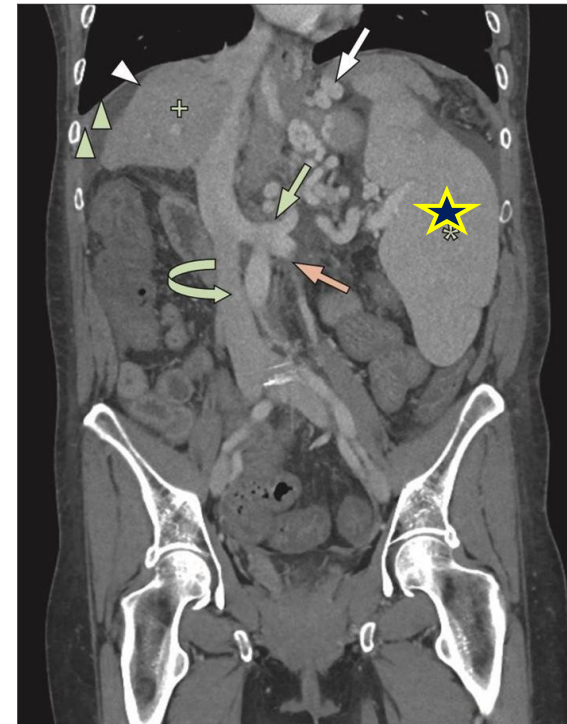
- A. Left gastric vein**
- B. Right gastric vein**
- C. Common hepatic vein**
- D. Inferior mesenteric vein**
- E. Superior mesenteric vein**



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SOM.MK.1.BPM2.3.DM.1.ANAT.1182 Describe the anatomical basis for the clinical symptoms of portal hypertension including esophageal varices, rectal varices and caput medusa



A 52-year-old woman comes to the physician because of a 7-month history of postprandial (after meals) epigastric pain, nausea, fullness and bloating. He says the symptoms started 1-month after starting a new medication. Medical history is significant for a 10-year history of Diabetes Mellitus Type 2. Gastric emptying studies show delayed gastric emptying, and a diagnosis of gastroparesis is made. The physician explains that damage to the parasympathetic nerves of the stomach caused the delayed gastric emptying. Patient saw improvement of symptoms following cessation of medication. Branches of which of the following nerves was most likely affected?

- A. Greater splanchnic**
- B. Lesser splanchnic**
- C. Least splanchnic**
- D. Pelvic splanchnic**
- E. Vagus**

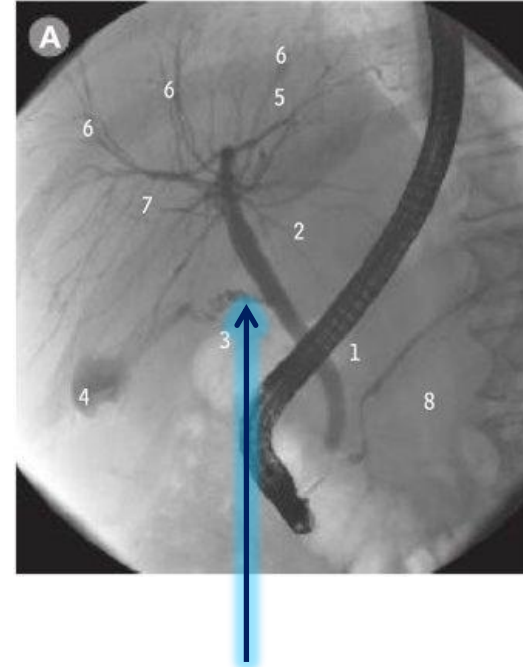
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SOM.MK.1.BPM2.3.DM.1.ANAT.1113Describe the sympathetic, parasympathetic and sensory innervation of the foregut, midgut and hind gut.

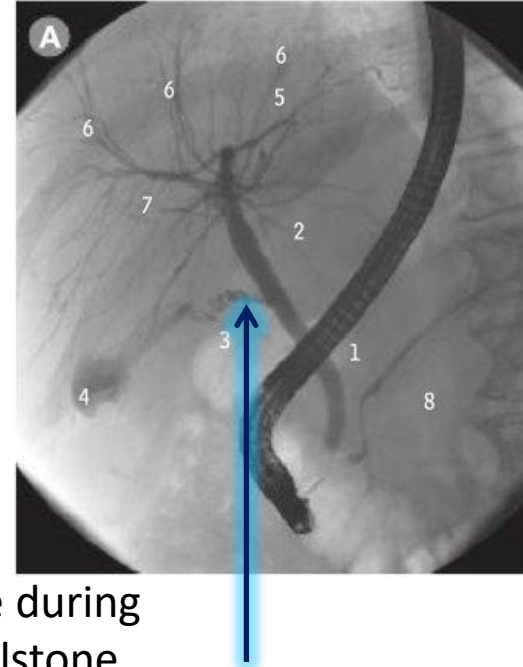
A 44-year-old woman comes to the emergency department because of a one-week history of right upper quadrant abdominal pain. She has recurrent, moderately intense, colicky pain every time she has a fatty meal. Her stool is pale-looking and urine is dark-colored. She has a history of hypertension and hyperlipidemia. Physical examination shows a woman with a BMI of 32 kg/m^2 , a low-grade fever, and tenderness to palpation in the right upper quadrant. ERCP imaging shows gallstones lodged in the highlighted portion of the biliary tree (see attached). Which of the following structures will most likely be affected?

- A. Gallbladder**
- B. Ampulla of Vater**
- C. Pancreatic Duct**
- D. Common Bile Duct**
- E. Common Hepatic Duct**



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- ☐ **D. Common Bile Duct**
- ☐ **E. Common Hepatic Duct**



SOM.MK.1.BPM2.3.DM.1.ANAT.1180

Explain the pathway taken by an endoscope from mouth to the biliary tree during Endoscopic Retrograde Cholangio Pancreatography (ERCP) to remove a gallstone

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Biochemistry 9 Questions

A 25-year-old man comes to the physician because of intermittent jaundice for the past several months. Physical examination shows scleral icterus. Laboratory findings are shown. Urinalysis is positive for bilirubin and negative for urobilinogen. A liver biopsy specimen shows dark, granular pigments in the hepatocytes. Which of the following best explains the clinical features in this patient?

	Patient	Reference
ALT	28 U/L	10-40 U/L
AST	25 U/L	12-38 U/L
ALP	61 U/L	25-100 U/L
Serum bilirubin total // direct	3.2 mg/dL // 2.8 mg/dL	0.1-1 mg/dL// 0.0-0.3 mg/dL

- A. Low hepatic UDP-glucuronyl transferase
- B. Oxidative stress induced hemolysis
- C. Impaired bilirubin secretion into the biliary system
- D. Copper accumulation in hepatocytes
- E. Impaired uptake of bilirubin into hepatocytes
- F. Hemosiderin accumulation in hepatocytes

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A 25-year-old woman comes to the physician because of a 10-day history of episodic severe abdominal pain, constipation, and weakness. She says that her urine was dark red and she thought it was blood. She has tachycardia and is agitated. Physical examination shows no bruising or other skin lesions. Urinalysis by dipstick shows no blood. Urine porphobilin level is 152 $\mu\text{mol/L}$ (normal $<10 \mu\text{mol/L}$). Which of the following best explains the accumulation of this metabolite in this patient?

- A. Mutation of Uroporphyrinogen-III synthase gene (*UROS*)
- B. Deficiency of uroporphyrinogen-III decarboxylase
- C. Heme deficiency stimulates ALAS1 expression
- D. Deficiency of ferrochelatase
- E. Oxidation of porphyrinogens
- F. Pyridoxine deficiency
- G. Inhibition of heme synthesis by lead

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A 4-month-old boy is referred for evaluation after he presented with focal seizures. Physical examination shows generalized hypotonia, lack of eye contact, dry skin and twisted hypopigmented hair. Cognitive examination shows decreased interaction with the family, inconstant smiling, no interest in toys or faces, and irritability. Genetic testing shows a mutation in the *ATP7A* gene. Which of the following enzymes would most likely be affected by the systemic lack of the nutrient transported by this gene product?

- A. Prolyl hydroxylase
- B. Lysyl hydroxylase
- C. Tyrosinase
- D. Glucose 6-phosphate dehydrogenase
- E. Branched chain alpha-keto acid dehydrogenase
- F. Methylmalonyl CoA mutase

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A 54-year-old woman comes to the physician because of a 2-day history of jaundice, fever and abdominal pain. Physical examination shows scleral icterus and right upper quadrant tenderness. Abdominal ultrasonography shows a shrunken gallbladder with cholelithiasis and an impacted large gallstone in the common bile duct with dilated common and intrahepatic bile ducts. Which of the following additional findings is most likely in this patient?

- A. Increased conjugated bilirubin secretion into the intestine
- B. Normal levels of stercobilin in feces
- C. Increased formation of urobilinogen in the intestine
- D. Increased serum conjugated bilirubin, due to regurgitation
- E. Decreased serum indirect bilirubin
- F. Normal serum alkaline phosphatase (ALP) levels
- G. AST: ALT ratio of greater than 2
- H. Low serum albumin levels

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A 26-year-old woman is brought to the emergency department by her friends because of a suspected overdose after being found semi-conscious with an empty bottle of acetaminophen. Her vital signs are within normal limits. Physical examination shows a drowsy patient unable to provide coherent answers. Laboratory studies are requested including liver function tests and she is empirically administered an antidote to her suspected overdose. Which of the following best explains the rationale behind the administration of the antidote in this patient?

- A. Inhibition of cytochrome P450 3A
- B. Increased conjugation with sulfate
- C. Increased conjugation with glucuronate
- D. Supply cysteine for glutathione replenishment
- E. Upregulation of cytochrome P450 2E1
- F. Inhibition of aldehyde dehydrogenase
- G. Inhibition of alcohol dehydrogenase

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A 69-year-old woman with alcohol use disorder is brought to the emergency department with progressive disorientation and disorganized speech. She has a 2-week history of watery diarrhea. Her diet consists only of corn flour bread. Her body mass index is 16.26 kg/m^2 (19-25). Her blood pressure is low. Physical examination shows dehydration, pallor, lethargy, disorientation, and loss of memory. She has a symmetrical rash with hyperkeratinization and hyperpigmentation on her neck, arms and legs. A diagnosis of nutritional vitamin deficiency is made. Which of the following reactions is most likely impaired in this patient?

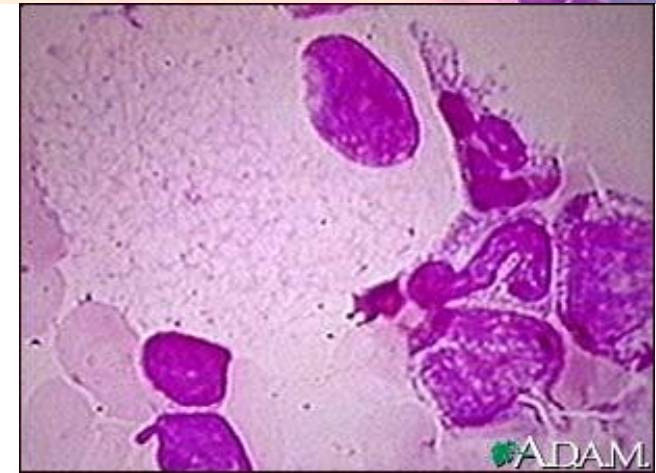
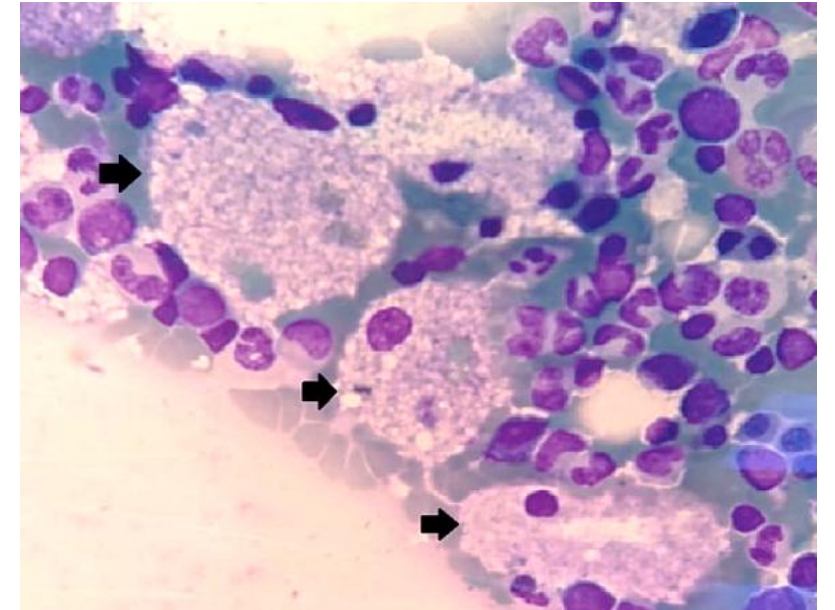
- A. Gamma-carboxylation of inactive blood clotting factors
- B. Conversion of glyceraldehyde 3-phosphate to 1,3-bisphosphoglycerate
- C. Formation of histamine by histidine decarboxylase
- D. Conversion of propionyl CoA to methylmalonyl CoA
- E. Formation of cysteine from homocysteine
- F. Non-oxidative sugar interconversion in pentose phosphate pathway
- G. Formation of hydrogen bonds in collagen

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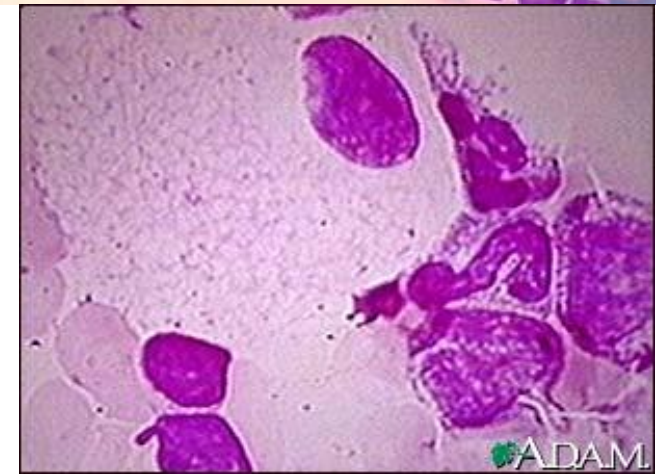
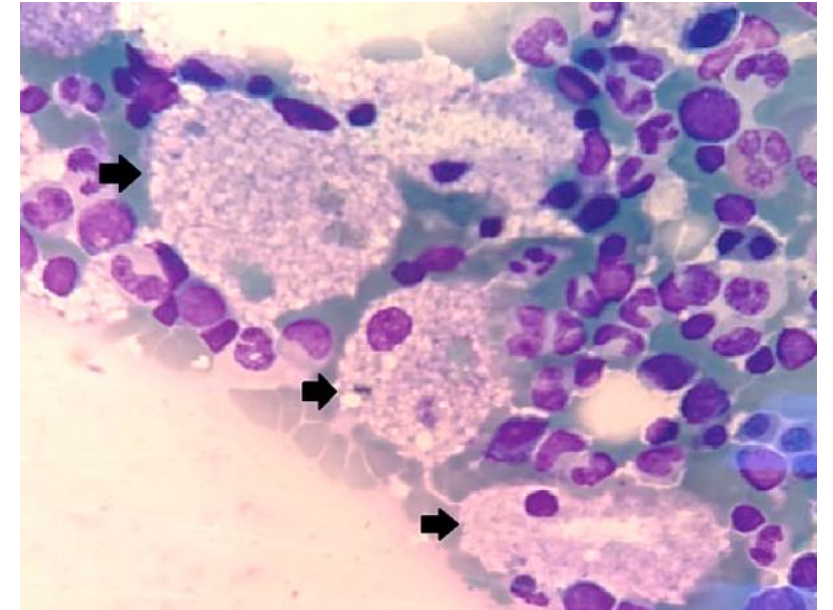
A 4-year-old boy is brought to the physician by his parents for a follow-up examination. He is 93 cm (3 ft 1 in) tall, weighs 15.2 kg (33.5 lb); his anthropometric parameters below the expected ranges). Physical examination shows nontender abdominal distension, the liver is palpable 3 cm below the right costal margin and the tip of the spleen is 10 cm below the left costal margin. A bone marrow biopsy is obtained, and the results shown in the photomicrographs. Which of the following is the most likely diagnosis for this patient?

- A. Niemann-Pick disease
- B. Tay-Sachs disease
- C. Hunter syndrome
- D. Fabry disease
- E. Gaucher disease
- F. I-cell disease
- G. Pompe disease



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SOM. MK.I.BPM2.3.DM.3.BCHM.1335; Distinguish between lysosomal storage diseases based on deficient enzyme, accumulating compound, clinical features, general cell appearance and clinical features: Hunter vs Hurler syndrome; Tay-Sachs disease : onion-shell inclusions & cherry-red macula; Fabry disease; Gaucher disease; Niemann-Pick disease: foamy appearing cells; Metachromatic leukodystrophy: accumulating compound; I-cell disease: Mannose-6-P marker for lysosomal enzymes

A 48-year-old man comes to the physician because of a 5-month history of fatigue, joint pain, and decreased libido. He was recently diagnosed with type 2 diabetes mellitus. Physical examination shows skin hyperpigmentation and swelling and tenderness over the metacarpophalangeal joints bilaterally. Abdominal examination shows hepatomegaly without abdominal distension. Which of the following laboratory findings is most likely decreased in this patient?

- A. Serum iron
- B. Serum hemoglobin
- C. Total iron binding capacity
- D. Serum ferritin
- E. Serum ceruloplasmin

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SOM. MK.III.BPM2.3.DM.3.BCHM.1341; Interpret significance of estimation of serum ferritin, serum iron, transferrin saturation and TIBC in iron deficiency anemia and hereditary hemochromatosis

SOM. MK.III.BPM2.3.DM.3.BCHM.1342; Discuss laboratory findings, inheritance and molecular basis of hereditary Hemochromatosis.

A 9-year-old boy is brought to the emergency department because of a 1-hour history of accidental ingestion of a clear, odorless liquid. He has nausea and vomiting. His father says the bottle contained ethylene glycol. Nasogastric aspiration is performed and fomepizole infusion is prescribed. Which of the following best describes the action of the prescribed drug?

- A. Activator of alcohol dehydrogenase
- B. Competitive inhibitor of alcohol dehydrogenase
- C. Allosteric inhibitor of alcohol dehydrogenase
- D. Non-competitive inhibitor of alcohol dehydrogenase
- E. Activates UDP-glucuronosyl transferase

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